

Student 32 **ROHIT KUMAR SEKAR** 192312214RA

4 / 4 submitted

Q1. Arrays, Array Applications and Structures

CODE

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[1000];
    if (scanf("%s", str) == 1) {
        int freq[256] = {0};
        int len = strlen(str);
        for (int i = 0; i < len; i++) {
            freq[(unsigned char)str[i]]++;
        }
        int found = 0;
        for (int i = 0; i < len; i++) {
            if (freq[(unsigned char)str[i]] == 1) {
                printf("%c\n", str[i]);
                found = 1;
                break;
            }
        }
        if (!found) {
            printf("-1\n");
        }
    }
    return 0;
}
```

INPUT `swiss`OUTPUT `(no output)`

Q2. Arrays, Array Applications and Structures

CODE

```

#include <stdio.h>

int main()
{
    int a[100], n, x, l=0, r, m;
    scanf("%d", &n);

    for(int i=0; i<n; i++) scanf("%d", &a[i]);
    scanf("%d", &x);
    r=n-1;

    while(l<r)
    {
        m=(l+r)/2;
        if(a[m]==x) { printf("%d", m); return 0; }

        if(a[l]<a[m])
            if(x<a[l] || x>a[m]) r=m-1;
            else l=m+1;
        else
            if(x>a[m] || x<a[r]) l=m+1;
            else r=m-1;
    }

    printf("-1");
}

```

INPUT

```

7
4 5 6 7 8 9 10
8

```

OUTPUT

(no output)

Q3. Arrays, Array Applications and Structures

CODE

```

#include <stdio.h>
int main()
{
    int a[100][2], n, i, j, t;
    scanf("%d", &n);
    for(i=0; i<n; i++)
        scanf("%d%d", &a[i][0], &a[i][1]);
    for(i=0; i<n-1; i++)
        for(j=i+1; j<n; j++)
            if(a[i][0]>a[j][0])
            {
                t=a[i][0]; a[i][0]=a[j][0]; a[j][0]=t;
                t=a[i][1]; a[i][1]=a[j][1]; a[j][1]=t;
            }
    for(i=0; i<n; i++)
    {
        if(i<n-1 && a[i][1]!=a[i+1][1])
        {
            if(a[i][1]<a[i+1][1]) a[i][1]=a[i+1][1];
            i++;
        }
        printf("%d %d\n", a[i][0], a[i][1]);
    }
}

```

INPUT

```

4
1 3
2 0
8 10
9 12

```

OUTPUT

(no output)

Q4. Arrays, Array Applications and Structures

CODE

```
#include <stdio.h>
#include <string.h>
int main() {
    char str[1000];
    if (scanf("%s", str) == 1) {
        int freq[256] = {0};
        int len = strlen(str);
        for (int i = 0; i < len; i++) {
            freq[(unsigned char)str[i]]++;
        }
        for (int i = 0; i < len - 1; i++) {
            for (int j = i + 1; j < len; j++) {
                unsigned char c1 = str[i], c2 = str[j];
                if (freq[c1] < freq[c2] || (freq[c1] == freq[c2] && c1 > c2)) {
                    char temp = str[i];
                    str[i] = str[j];
                    str[j] = temp;
                }
            }
        }
        printf("%s", str);
    }
    return 0;
}
```

INPUT tree

OUTPUT (no output)